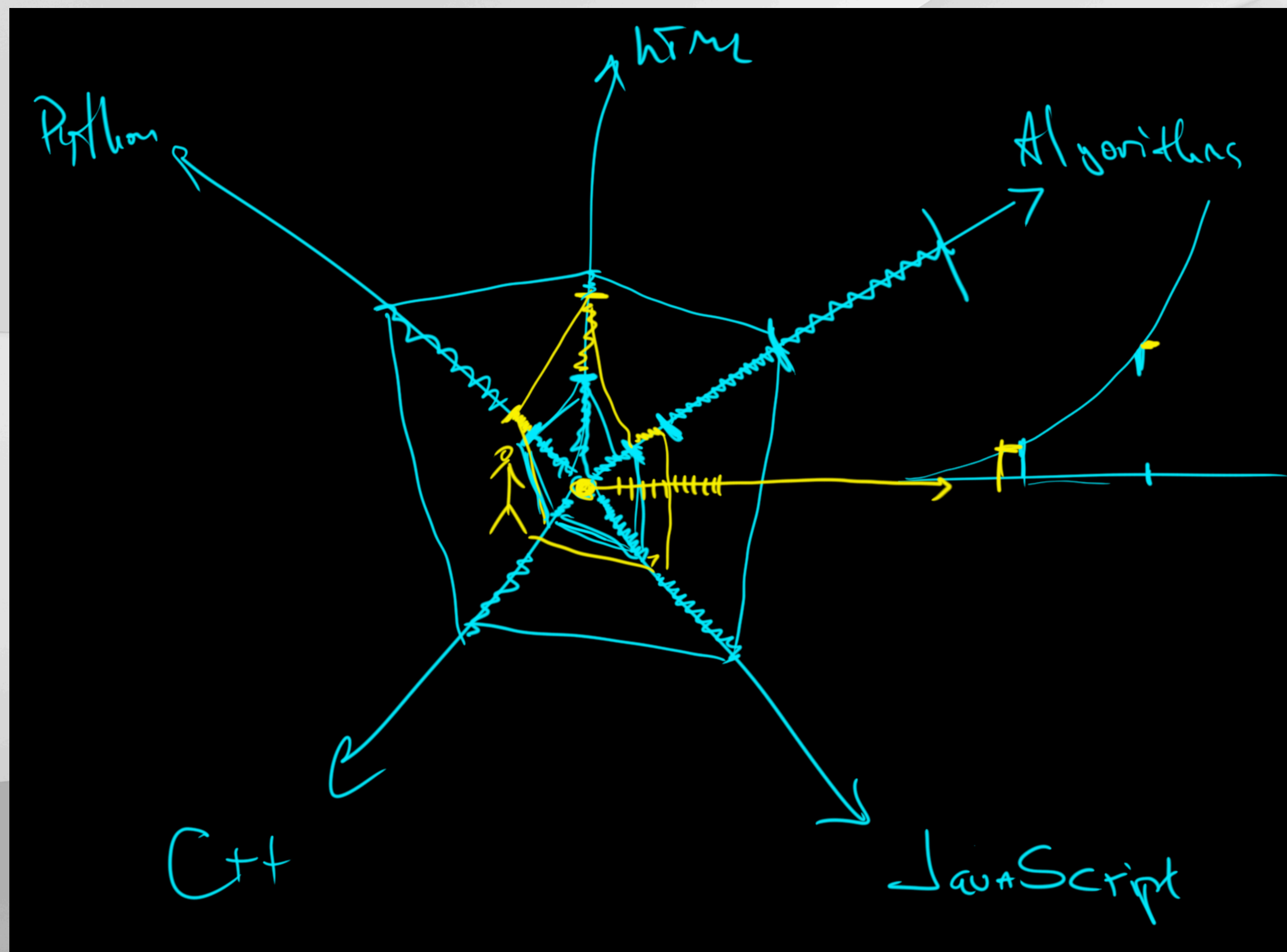
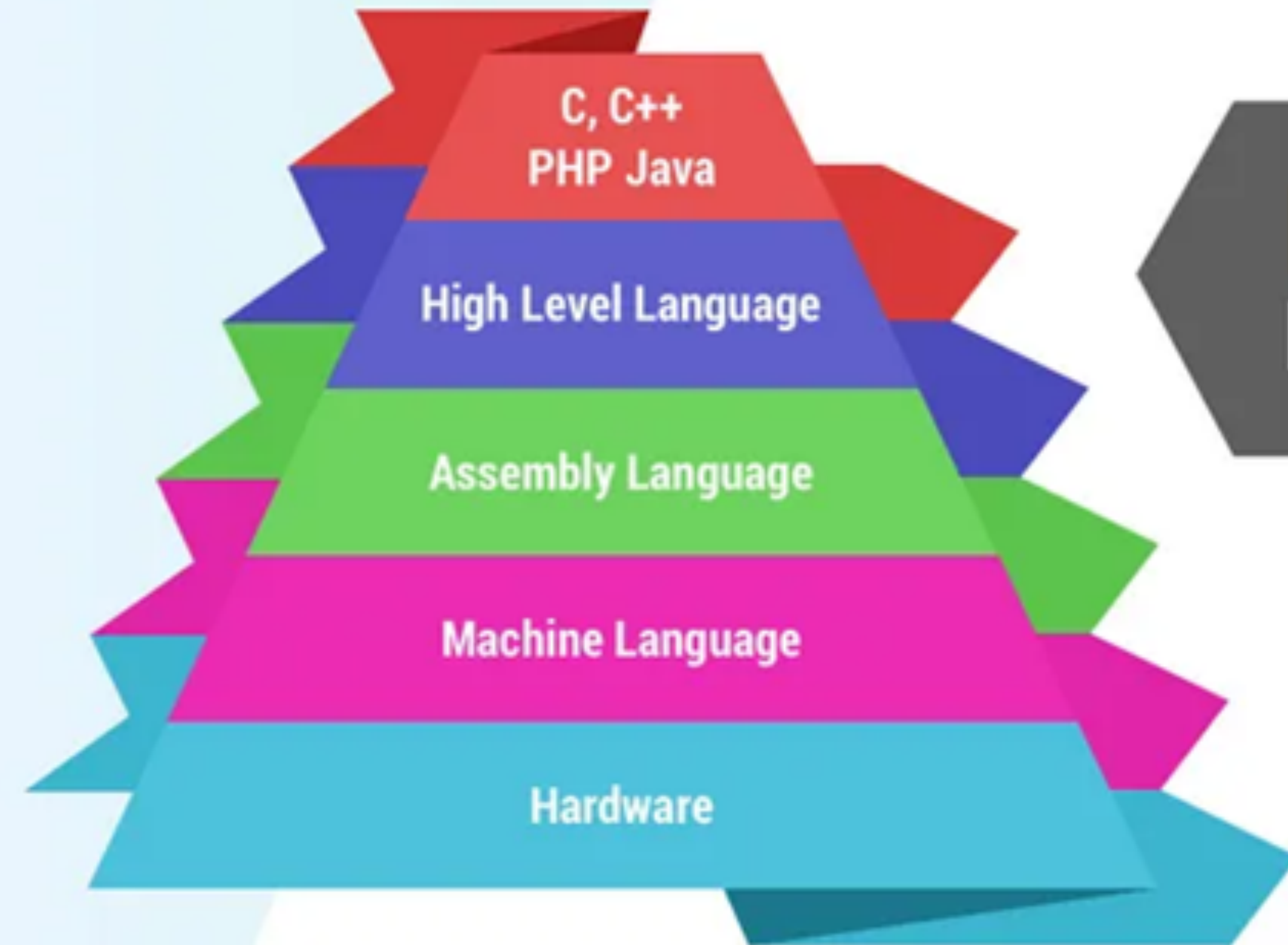


Низькорівневе програмування та програмування мікроконтролерів

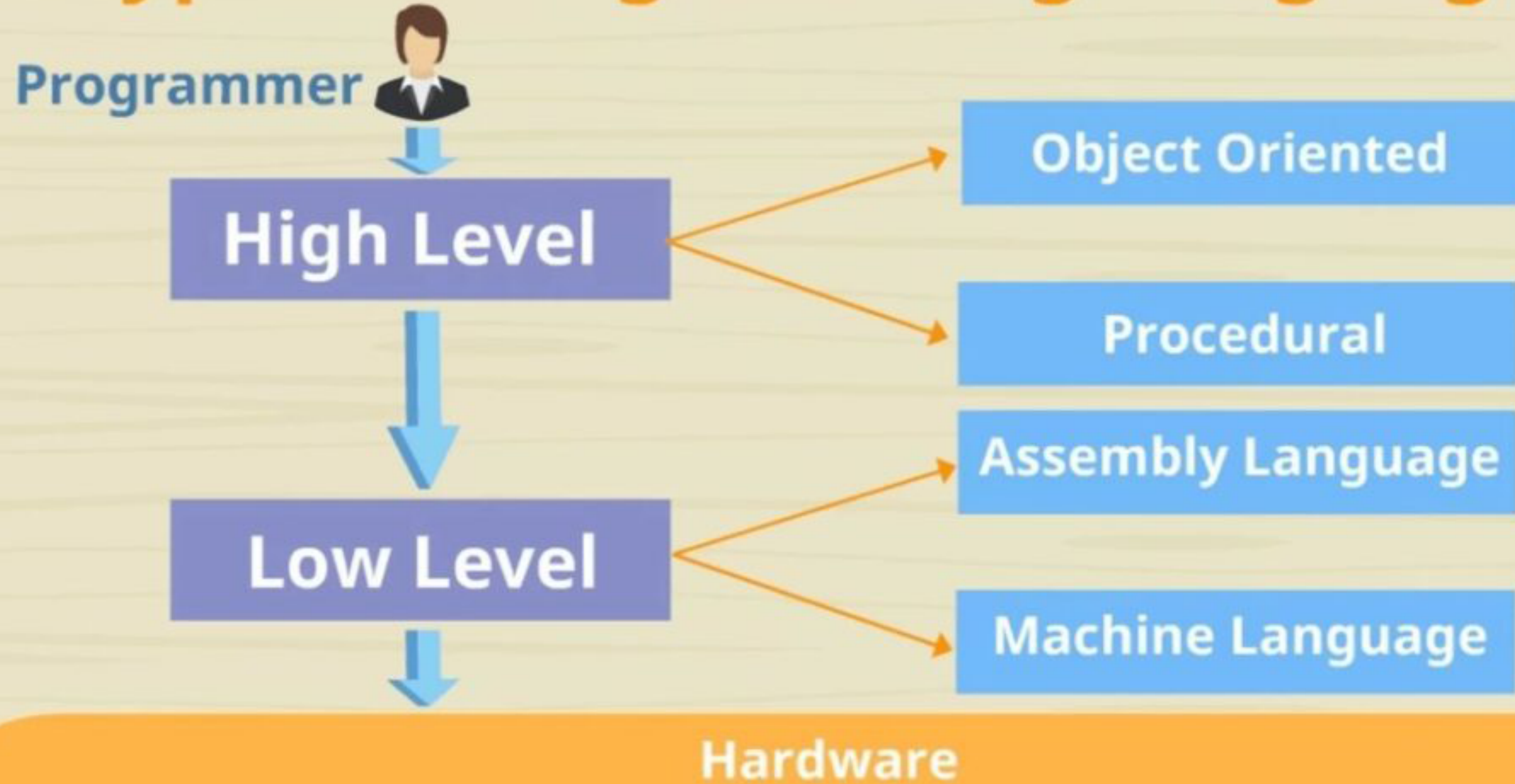
З використанням мови програмування **Assembler**





LEVELS OF PROGRAMMING

Types of Programming Languages




```

C000                                ORG    ROM+$0000 BEGIN MONITOR
C000 8E 00 70  START  LDS    #STACK

*****
* FUNCTION: INITA - Initialize ACIA
* INPUT: none
* OUTPUT: none
* CALLS: none
* DESTROYS: acc A

0013                                RESETA EQU    %00010011
0011                                CTLREG EQU    %00010001

C003 86 13  INITA  LDA A  #RESETA  RESET ACIA
C005 B7 80 04                                STA A  ACIA
C008 86 11                                LDA A  #CTLREG  SET 8 BITS AND 2 STOP
C00A B7 80 04                                STA A  ACIA

C00D 7E C0 F1                                JMP    SIGNON  GO TO START OF MONITOR

*****
* FUNCTION: INCH - Input character
* INPUT: none
* OUTPUT: char in acc A
* DESTROYS: acc A
* CALLS: none
* DESCRIPTION: Gets 1 character from terminal

C010 B6 80 04  INCH  LDA A  ACIA      GET STATUS
C013 47                                ASR A      SHIFT RDRF FLAG INTO CAR
C014 24 FA                                BCC    INCH  RECIEVE NOT READY
C016 B6 80 05                                LDA A  ACIA+1  GET CHAR
C019 84 7F                                AND A  #$7F    MASK PARITY
C01B 7E C0 79                                JMP    OUTCH  ECHO & RTS

*****
* FUNCTION: INHEX - INPUT HEX DIGIT
* INPUT: none
* OUTPUT: Digit in acc A
* CALLS: INCH
* DESTROYS: acc A
* Returns to monitor if not HEX input

C01E 8D F0  INHEX  BSR    INCH      GET A CHAR
C020 81 30                                CMP A  #'0    ZERO
C022 2B 11                                BMI    HEXERR NOT HEX
C024 81 39                                CMP A  #'9    NINE
C026 2F 0A                                BLE    HEXRTS GOOD HEX
C028 81 41                                CMP A  #'A
C02A 2B 09                                BMI    HEXERR NOT HEX
C02C 81 46                                CMP A  #'F
C02E 2E 05                                BGT    HEXERR
C030 80 07                                SUB A  #7     FIX A-F
C032 84 0F  HEXRTS AND A  #$0F    CONVERT ASCII TO DIGIT
C034 39                                RTS

```